SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXP:15

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Test Plan Document for an Inventory Management System

Aim

To develop a comprehensive Test Plan for an Inventory Management System, covering the test scope, test strategy, test design techniques, test execution strategy, environment, schedule and test completion criteria.

Problem Statement

An inventory management system tracks products, stock levels, suppliers, purchase orders and stock movement between warehouses. Wrong stock counts cause shortages, over-stocking and financial loss, so the system needs careful and well-planned testing.

Testing is needed to check that the application:

- Product, supplier and stock records are managed correctly.
- Stock-in, stock-out, transfer and adjustment update the quantities correctly.
- Alerts, reports, access control and backup work as specified.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	Users log in with role-based access (admin, store manager, staff).
FR2	Users can add, edit and delete products with a unique SKU, name, category, unit price and reorder level.
FR3	Stock-in is recorded through a purchase order and a goods received note (GRN).
FR4	Stock-out is recorded through sales or issue and cannot exceed the available stock.
FR5	An alert is raised when the stock falls to the reorder level.
FR6	The admin can add, update and deactivate suppliers.
FR7	A purchase order must be approved by the store manager before it is sent to the supplier.
FR8	A stock adjustment needs a reason and is saved in an audit log.
FR9	Items can be scanned by barcode for stock-in, stock-out and search.
FR10	Stock can be transferred between warehouses, and the balance of both warehouses is updated.
FR11	The system gives reports for stock valuation, stock movement and low stock.
FR12	Data is backed up every day and can be restored.

2 Software and environment requirements

Item	Details
Application	Web application (Java/Node.js back end, Angular front end) with a barcode scanner interface
Database	MySQL / PostgreSQL (products, stock, suppliers, orders, audit log)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, JMeter, bug tracker (Jira)
Test data	100 products, 5 suppliers, 2 warehouses, sample purchase orders, barcode labels

Test Plan

1 Test plan details

Item	Details
Project	Inventory Management System
Document	Test Plan
Version	1.0
Prepared by	Balaguru S (192312070)
Date	01-11-2026
Reference documents	Software Requirements Specification (functional requirements FR1 to FR12 above)

2 Test objectives

- Verify that every functional requirement is implemented correctly and can be traced to at least one test case.
- Find defects early and report them with enough detail to fix them.
- Check non-functional qualities: usability, performance, security and compatibility.
- Provide clear completion criteria so that the release decision is based on test results.

3 Test scope

In scope:

- Login and role-based access control.
- Product, supplier and purchase order management.
- Stock-in, stock-out, adjustment and warehouse transfer.
- Reorder alerts and stock reports.

- Barcode scanning and audit log.
- Performance with large stock data, backup and restore.

Out of scope:

- Accounting and payroll modules.
- Supplier's own systems and delivery vehicles.
- Physical stock counting at the warehouse.
- Hardware testing of barcode scanners.

4 Test strategy and test levels

Test level	Focus	Performed by	Tools / method
Unit testing	Each function or class works on its own	Developers	JUnit / NUnit, code review
Integration testing	Modules work together (UI, business logic, database)	Developers and testers	Postman, API tests
System testing	Complete application against all requirements	Test team	Manual tests, Selenium
Regression testing	Old features still work after changes or fixes	Test team	Automated Selenium suite
Non-functional testing	Performance, security, usability, compatibility	Test team	JMeter, browser checks
User acceptance testing	Users confirm the system meets their needs	End users and client	Business scenarios

5 Test design techniques

Technique	Applied to	Example
Equivalence partitioning	Product data, quantity, user roles	Valid SKU, duplicate SKU, blank SKU; admin, manager, staff
Boundary value analysis	Stock-out, reorder level	Issue stock equal to, one below and one above the available quantity
Decision table	Purchase order approval	Combinations of order value, approval status and supplier status
State transition	Purchase order status	Draft → Approved → Sent → Received → Closed
Use case testing	End-to-end stock cycle	Create PO → receive goods → issue stock → run report
Error guessing	Quantities and forms	Negative quantity, decimal quantity, very large numbers

Requirement-to-test coverage:

Req. ID	Test condition	Technique
FR1	Login with each role; access to restricted pages by staff	Equivalence partitioning
FR2	Add a product with a unique and a duplicate SKU; edit and delete a product	Equivalence partitioning
FR3	Raise a purchase order, receive goods with a GRN and check the stock	Use case testing
FR4	Issue stock equal to, below and above the available quantity	Boundary value analysis
FR5	Reduce stock to the reorder level and check the alert	Boundary value analysis
FR6	Add, update and deactivate a supplier; use a deactivated supplier in a PO	Decision table
FR7	Send an unapproved PO; approve and send a PO	Decision table
FR8	Adjust stock with and without a reason; check the audit log	Error guessing
FR9	Scan a valid, an invalid and a damaged barcode	Equivalence partitioning
FR10	Transfer stock between warehouses; transfer more than available	Boundary value analysis
FR11	Generate each report and compare totals with the database	Use case testing
FR12	Take a backup, delete test data and restore it	State transition

6 Test execution strategy

Testing is carried out in cycles. Each cycle starts when the build meets the entry criteria and ends when the cycle goals are reached.

Cycle	Tests executed	Goal
Cycle 1: Smoke	Login and the most important flow of the application	Confirm the build is stable enough for detailed testing
Cycle 2: Functional	All positive and negative test cases for FR1 to FR12	Find functional defects and report them
Cycle 3: Regression	Re-test fixed defects and run the regression suite	Confirm fixes and make sure nothing else is broken
Cycle 4: Non-functional	Performance, security, usability and compatibility tests	Check quality attributes against the targets

Cycle	Tests executed	Goal
Cycle 5: UAT	Business scenarios run by end users	Get user sign-off for release

Execution rules:

- Test cases are executed in priority order: high, medium, low.
- Each result (Pass, Fail, Blocked) is recorded, and every failure is logged as a defect with steps, expected and actual result.
- Fixed defects are re-tested, and the related areas are regression tested.
- A daily status report shows the test cases run, passed, failed and blocked, and the open defects.

7 Test environment

Item	Details
Application	Web application (Java/Node.js back end, Angular front end) with a barcode scanner interface
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8 Roles and responsibilities

Role	Responsibility
Test manager	Prepares the test plan, tracks progress, reports to stakeholders
Test engineer	Writes and runs test cases, logs defects, re-tests fixes
Automation engineer	Builds and maintains the automated regression suite
Developer	Fixes defects and supports unit and integration testing
Business analyst / client	Clarifies requirements and gives UAT sign-off

9 Test schedule

Activity	Duration	Responsible
Test planning and requirement review	1 week	Test manager
Test scenario and test case design	2 weeks	Test engineers

Activity	Duration	Responsible
Environment and test data setup	1 week	Test team, developers
Cycles 1 and 2: smoke and functional testing	3 weeks	Test engineers
Cycle 3: regression testing	1 week	Test and automation engineers
Cycle 4: non-functional testing	1 week	Test team
Cycle 5: user acceptance testing	1 week	End users, test manager
Test summary report and sign-off	2 days	Test manager

10 Test completion criteria

Entry criteria:

- The requirements are approved and the test cases are reviewed.
- The build is deployed in the test environment and passes the smoke test.
- Test data and test accounts are ready.

Exit (completion) criteria:

- 100% of the planned test cases are executed.
- At least 95% of the executed test cases pass.
- No Critical or High severity defect is open.
- All functional requirements are covered by at least one executed test case.
- The regression suite passes on the final build, and the UAT sign-off is received.

Suspension and resumption:

- Testing is suspended if the build is not stable (for example, login fails) or the test environment is down.
- Testing resumes when a new build passes the smoke test or the environment is restored.

11 Risks and mitigation

Risk	Likelihood	Impact	Mitigation
Wrong stock quantities because of simultaneous updates	Medium	High	Run concurrency tests with several users issuing the same item
Large data volume slows reports	Medium	Medium	Run performance tests with realistic data volume
Barcode scanner not available for testing	Medium	Low	Use scanner simulation or manual barcode entry
Requirement changes late in the project	Medium	Medium	Keep requirement-to-test coverage updated and run regression tests

12 Defect management

Severity	Meaning	Target action
Critical	System crash, data loss or security breach; no workaround	Fix before the next build
High	Major feature fails; no easy workaround	Fix in the current cycle
Medium	Feature partly fails; a workaround exists	Fix before release
Low	Cosmetic or minor issue	Fix if time permits

Defect life cycle: New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still present after retesting, it is reopened and sent back to the developer.

13 Test deliverables

- Test plan (this document)
- Test scenarios and test cases, with the requirement traceability matrix
- Test data and test environment details
- Defect reports and daily status reports
- Test summary report

Result / Conclusion

Result: A complete test plan was prepared for the Inventory Management System. It covers the test scope, 6 test levels, 6 test design techniques, a requirement-to-test coverage table for all 12 functional requirements, a 5-cycle execution strategy, the environment, roles, schedule, completion criteria, risks and defect management.

Conclusion: The test plan gives the Inventory Management System project a clear and repeatable way of testing. Testing against fixed completion criteria, with every requirement traced to tests, helps find defects early and supports a reliable release decision. The plan should be reviewed and updated whenever the requirements or the schedule change.